Tables and Figures

March 31, 2026

Table 1: MSEP, MSEC, MFD and inference time comparison with more baselines. TrackGPT and MENet are closed source and under implementation. We will update it before Apr. 7th.

metrics	MSEP	MSEC	MFD	time
our model	**0.41**	**1.23e-03**	**7.80**	**62**
MP-LSTM	1.6	0.00	31.11	287
TrAISformer	0.71	1.19e-02	19.78	541
SEMINT	2025.65	269.92	127.24	374
DiffuTraj	0.328	5.382	9.25	289
AISFuser	2863.503	2.47	29.62	163
TrackGPT				
MENet				

Table 2: Horizon-wise. MFD for each model under different prediction horizon from 12 points to 144 points. TrackGPT and MENet are closed source and under implementation. We will update it before Apr. 7th.

MFD	12	24	36	48	60	72	84	96	108	120	144
our model	0.59	1.04	1.50	2.51	3.41	4.36	4.71	5.13	5.85	6.78	7.80
MP-LSTM	6.41	12.04	16.69	20.30	22.86	24.44	24.90	25.67	24.90	28.47	31.11
TrAISformer	11.91	12.50	13.34	14.29	14.86	15.50	16.26	16.95	17.91	18.46	19.78
SEMINT	126.02	126.14	126.32	126.42	126.57	126.72	126.85	126.97	127.11	127.24	127.24
DiffuTraj	3.00	3.59	4.04	5.72	6.25	6.51	7.09	7.78	8.47	8.92	9.25
AISFuser	6.51	8.51	10.29	12.58	14.39	16.72	18.04	19.80	21.83	25.12	29.62
TrackGPT											
MENet											

Table 3: threshold test. Fallback to TOP-ONE if most voted NKPs are less than 50(total trajectory: 34058)

threshold	accuracy	fallback
0	0.4956	0
0.25	0.6640	0
0.5	0.7538	0
0.75	0.7965	62
0.98	0.8986	6956
top one	0.9361	34058

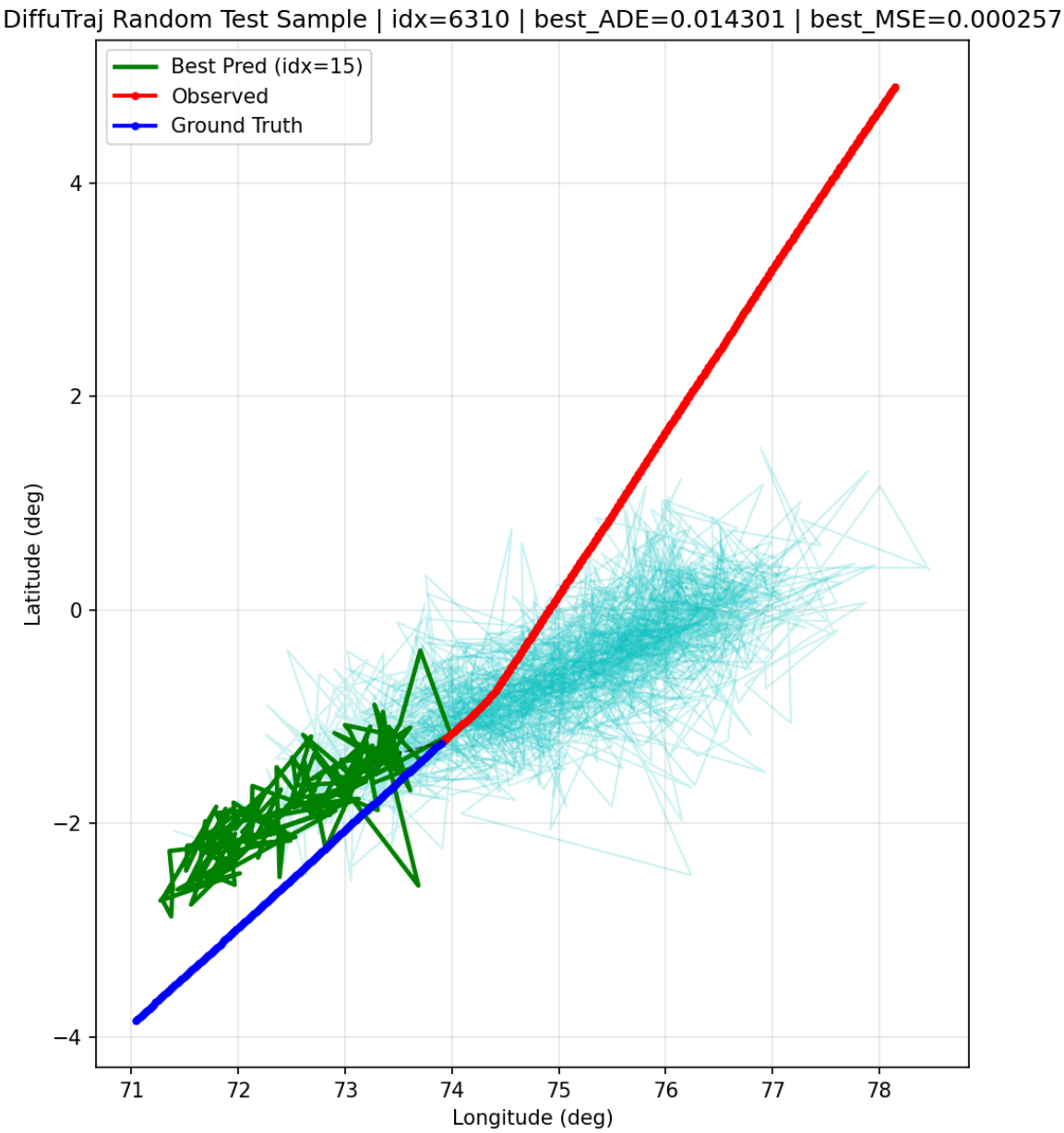

Figure 1: DiffuTraj Sample. It is the best, except for our model in metrics, but it has a large bias and a tendency to circle.